SECTION A

QUESTION 1 (compulsory)

(a) Explain various parameters used in classification and identification of plant diseases **(12 marks)**

(b) Explain how Fluorescence In-Situ Hybridization (FISH) works as a molecular method of disease identification. Further state the advantages of this molecular method. **(8 marks)**

SECTION B

QUESTION 2

(a) In details, discuss plant disease tetrahedron **(8 marks)**

(b) Define the following terms
i. Stridulation
ii. Phytopathogenesis
iii. Biotrophs
iv. Phytoanticipins **(8 marks)**

(b) Describe various distinct stages involved in pathogenesis **(4 marks)**

QUESTION 3

(a) Insects are living organisms that belong to the class *insecta* and phylum *athropoda*. State and describe five (5) *"Orders"* into which insects can be classified including examples of insects that belong to such *Orders*. **(10 marks)**

(b) Briefly explain how bacteria reproduce **(5 marks)**

(c) Describe the differences between viruses and viroids **(5 marks)**

ESTION 4

(a) Pests are known to develop resistance to a certain pesticide after a given period of time. Discuss how this phenomenon occur (5 marks)

(b) Explain how plants defend themselves against infections using pre-existing defense structures (10 marks)

(c) Nematodes are known to cause a number of problems and diseases in plants. Discuss the damages caused by nematodes and how they cause such damages (5 marks)

SECTION C

QUESTION 5

(a) Discuss similarities and differences between compound interest diseases and simple interest diseases. (10 marks)

(b) Briefly, explain how Polymerase Chain Reaction (PCR) works as a plant disease identification method. (5 marks)

(c) List various symptoms and signs caused by plant bacterial diseases. (5 marks)

QUESTION 6

(a) Discuss the four-tiered approach that anchors Integrated Pest Management (IPM). (8 marks)

(b) Explain in details how viruses invade the plant and cause diseases (4 marks)

(C) Briefly explain various characteristics of arthropods (8 marks)

QUESTION 7

(a) Explain various cultural practices that aim at reducing the adverse effects of plant diseases. (10 marks)

(b) In details, explain the imporiance of understanding insect classifications as an Agroforester (10 marks)

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PLANT PROTECTION (NR430) 2023 , TEST:1 TIME: 1hr: 30 min

Answer all the questions.

1. What do you understand by the following terms? Similar

   (a). Hemibiotroph (b). Disease syndrome (c).Necrotroph (d). Predisposition (8 marks)

2. State and explain different parameters used to identify and describe plant diseases (12 marks)

3. Describe various symptoms associated with necrosis in fungal disease infestation in plants (10 marks)